

Week 7 Module Project B: Q-Learning with Bayesian Belief Tracking

Acting under partial observability with normalized posterior state

CMPSC 480 - Instructor Solution Deck

Instructor material - do not distribute with the question deck

Solution overview

This instructor deck provides a complete matched solution and grading guidance.

Coverage: Weeks 4-7: MDPs, Q-learning, approximation, Bayes nets, and inference

Points

20 total

Expected time

10-14 hours

Submission

Repository, evaluation report,
and belief traces

Instructor verification checklist

- ▶ Use Python 3.11 and NumPy; do not use a library that implements the assessed algorithm.
- ▶ Keep data, model, optimization, and evaluation responsibilities behind explicit interfaces.
- ▶ Validate shapes, ranges, finite values, empty inputs, and terminal or degenerate cases.
- ▶ Use deterministic seeds and record every configuration used for reported evidence.
- ▶ Include focused tests and a README with exact reproduction commands.

Solution 1.a - Belief model (2 points)

Question focus. Write the vectorized posterior update.

Answer. For each hypothesis h , compute $\text{unnormalized}[h] = \text{likelihood}(\text{observation}|h, \text{agent_state}) * \text{prior}[h]$, then divide by the sum. Validate a nonnegative finite prior that sums to one.

Solution 1.b - Belief model (2 points)

Question focus. Hand compute prior $[0.5, 0.5]$ with likelihoods $[0.8, 0.2]$ for one observation.

Answer. Unnormalized masses are $[0.4, 0.1]$, normalization is 0.5, and posterior is $[0.8, 0.2]$.

Solution 1.c - Belief model (1 points)

Question focus. Define behavior when the normalization constant is zero or nonfinite.

Answer. Raise a domain-specific impossible-evidence error or apply a documented fallback such as restoring the prior; never silently divide by zero or invent a posterior.

Solution 2.a - RL agent (2 points)

Question focus. Define the observable agent state and prove it excludes the true hidden hazard.

Answer. The state may include position, local observation, and belief vector or its summary. A test inspects inputs and confirms the true hazard identifier is never passed to policy or update code.

Solution 2.b - RL agent (2 points)

Question focus. Specify epsilon-greedy selection and the terminal Q update.

Answer. Choose uniformly among legal actions with probability epsilon, otherwise stable argmax. For terminal steps target equals immediate reward; for others add gamma times max next Q.

Solution 3.a - Integration contract (2 points)

Question focus. State the within-step order from observation to action to learning.

Answer. Condition the prior on the new observation, build the policy state from the updated belief, choose a legal action, execute it, then apply the reward transition update under the documented convention.

Solution 3.b - Integration contract (1 points)

Question focus. Test that reset restores the declared prior and clears episode-local state.

Answer. Two episodes after reset begin with identical beliefs and agent state under the same seed, with no posterior or history leaked from the previous episode.

Solution 3.c - Integration contract (1 points)

Question focus. List every component needed to resume training exactly.

Answer. Save Q parameters or table, belief configuration, online belief if mid-episode, epsilon, counters, environment state if supported, and all random-generator states.

Solution 4.a - Noise study and ablation (2 points)

Question focus. Define safety and performance metrics and run multiple seeds.

Answer. Report mean return, success, hazard rate, and belief accuracy or entropy with variability for identical evaluation episodes and epsilon zero.

Solution 4.b - Noise study and ablation (2 points)

Question focus. Ablate the posterior input and interpret the difference.

Answer. Replace the belief with a fixed prior or observation-only state while holding all else fixed. Attribute only the controlled performance change to belief integration.

Solution 5.a - Defensive evidence (2 points)

Question focus. Provide four tests and name the defect each exposes.

Answer. Use repeated identical evidence, impossible evidence, uninformative sensor likelihoods, and a terminal transition; each test states expected belief or target and its associated bug.

Solution 5.b - Defensive evidence (1 points)

Question focus. Explain why a high-noise posterior should remain closer to the prior.

Answer. When likelihoods are similar across hypotheses, multiplying by them changes relative prior masses little; normalized posterior information gain is therefore small.

Edge-case reasoning and expected behavior

- ▶ Reject the prior.
- ▶ Raise or use the documented impossible-evidence fallback.
- ▶ Posterior equals prior.
- ▶ Return without bootstrapping.
- ▶ Fail the leakage test and remove the hidden value.

Grading rubric - part 1

Category	Points	Full-credit evidence
Belief model	5	Bayes numerator, normalization, repeated evidence, and validation are correct.
RL agent	4	No hidden-state leakage; exploration, targets, and termination are correct.
Integration contract	4	Update order, state encoding, reset, and episode isolation are correct.

Grading rubric - part 2

Category	Points	Full-credit evidence
Noise study and ablation	4	Noise, seeds, budgets, baselines, metrics, and uncertainty are reported.
Defensive evidence	3	Tests cover impossible evidence, high noise, ties, and normalization drift.

Reflection exemplar

Answer. A strong response verifies the posterior against an oracle, then uses visitation, reward design, policy state, or Q-learning evidence to locate the remaining control defect.

Grading guidance. Award credit for a specific claim supported by technical evidence from the submitted work.